Image Processing and Medical Imaging

Statistical Learning in Image Processing

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Some applications of statistical learning

- Medical: Predicted whether a patient, hospitalized due to a heart attack, will have a second heart attack. Risk factors for prostate cancer. Data: demographic, clinical measurements, etc.
- Business/Economics: Predict the price of stock 6 months from now. Data: company performance, economic data.
- Computer Vision: Identify handwritten numbers. Data: handwritten numbers

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Types of data

- Two basically different types of data
  - Numerical, e.g. stock price
  - Binary, e.g. cancer/no cancer
- Data are predicted
  - On the basis of a set of features
  - From a set of (observed) **training data** on these features
  - Inputs for the problems are also called predictors or independent variables.
  - Outputs are also called responses or dependent variables.

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Types of data (cont.)

- The prediction model is called a **learner** or **estimator**.
  - **Supervised learning**: learn on outcomes for observed features
  - **Unsupervised learning**: no feature values available

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Example: Email spam

- Data
  - Thousands of email messages each labeled + (email) or – (spam). This the training set.
  - The relative frequencies of the 57 most commonly occuring words and punctiation marks in the message.
- Prediction goal: label future messages (+) email, or (-) spam.
- Supervised learning problem on categorical data: classification problem
- Tolerance to errors:
  - Tolerant to letting through some spam (false positive)
  - No tolerance towards throwing out email (false negative)

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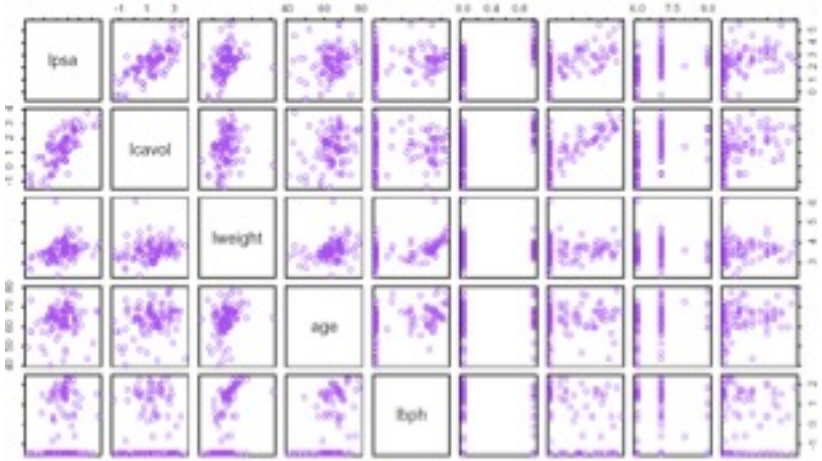
Example: Prostate cancer

- Data: cancer volume, prostate weight, age, etc.
- Predict: PSA (prostate specific antigen) level.
- Supervised learning problem on quantitative data: regression problem.



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Example: Prostate cancer (cont.)

- Figure shows scatter plots of the input data, projected onto two variables, respectively.
- The first row shows the outcome of the prediction, projected onto each input variable, respectively.

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Example: Recognition of handwritten numbers

- Data: images are single digits 16x16 8-bit gray-scale normalized for size and orientation.
- Classify: newly written digits
- Non-binary classification problem
- Low tolerance to misclassifications

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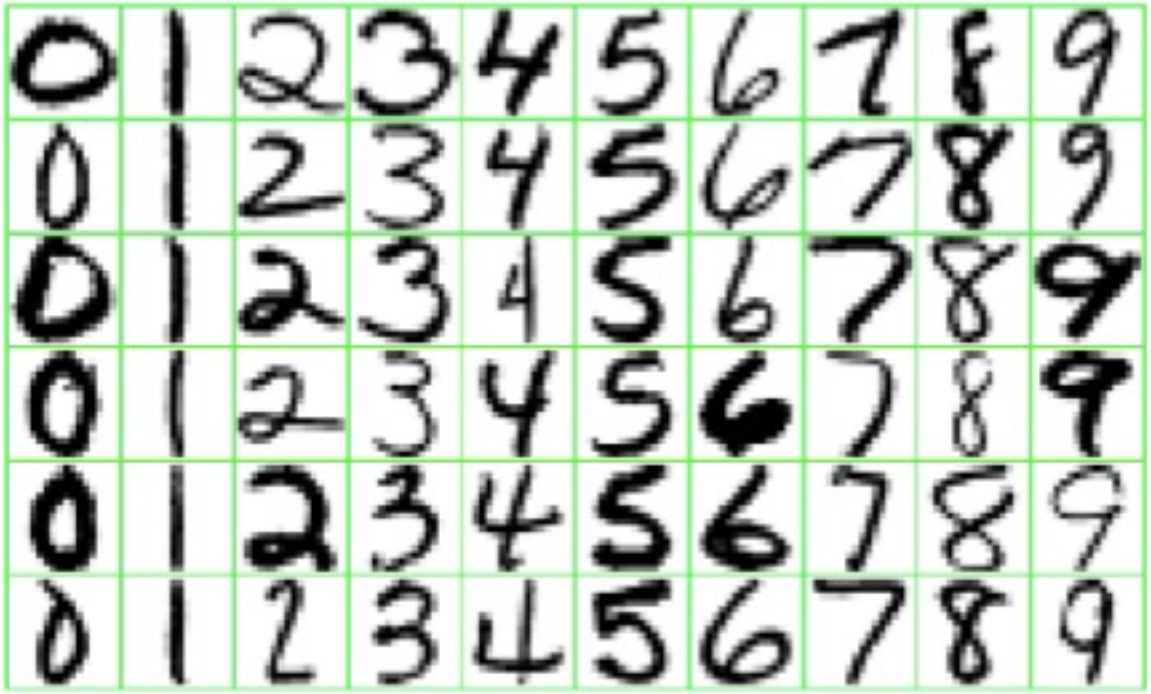
Example: Recognition of handwritten numbers



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Source: http://www.math.udel.edu/~dguillot/teaching/MATH829/lectures-handout/10-MATH829-categorical-handout.pdf

Supervised learning: Least-Squares

- Given inputs $\boldsymbol{X=(X_1,X_2,\ldots,X_l)}$. Predict output $\boldsymbol{Y}$ via the model

$$\hat{Y} = \hat{\beta}_0 + \sum_{j=1}^{p} X_j \hat{\beta}_j$$

- In the p+1-dimensional input-output space represents a hyperplane
- If the constant is included in X, the hyperplane goes through the origin.

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Supervised learning: Least-Squares (example)

- Data on two inputs X_1 and X_2.
- Output variable has values green (coded 0) and red (coded 1).
- 100 points per class
- Regression line is defined by $x^T\beta=0.5$
- Easy but many missclassifications, if the problem is not linear.

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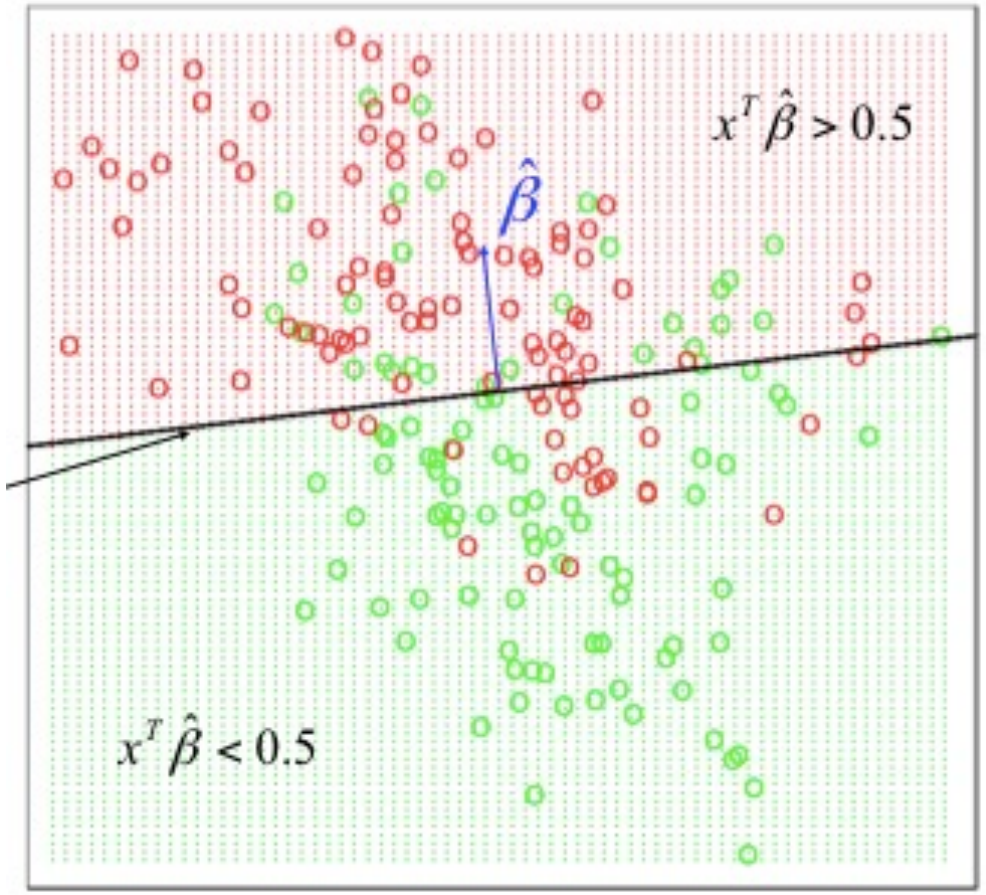
Sorce: https://www.isegi.unl.pt/docentes/vlobo/isegi_dm2/DM2_3_Bayes.pdf

Supervised learning: Least-Squares (example)



Source: https://www.isegi.unl.pt/docentes/vlobo/isegi_dm2/DM2_3_Bayes.pdf

Supervised learning: Nearest neighbors

- Uses those observations in the training set closest to the given input
- $N_k(x)$ is the set of the k closest points to x is the training sample.
- Average the outcome of the k closest training sample points.
- Fewer misclassifications.

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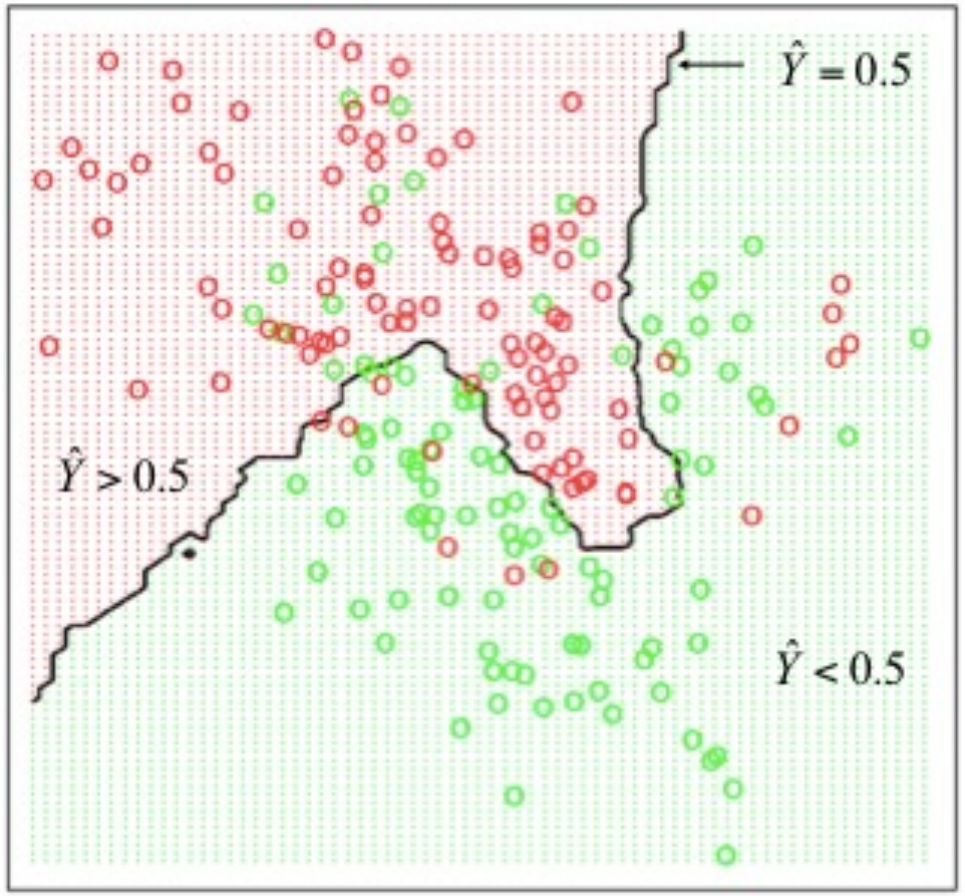
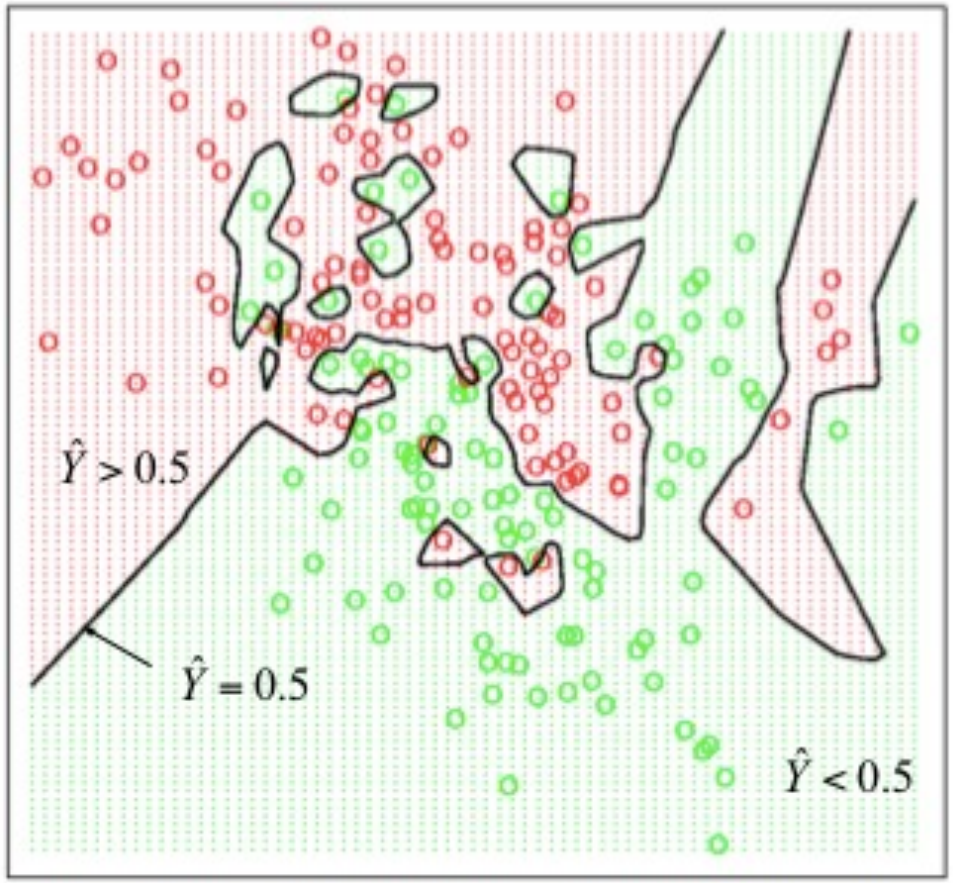
Supervised learning: Nearest neighbors (example)



Source: https://www.isegi.unl.pt/docentes/vlobo/isegi_dm2/DM2_3_Bayes.pdf

Supervised learning: Nearest neighbors (overfitting)



Source: https://www.isegi.unl.pt/docentes/vlobo/isegi_dm2/DM2_3_Bayes.pdf

Comparison of these approaches

- Least Squares
  - *p* paramters, *p* features
  - low variance (robust)
  - high bias
- K-nearest neighbors
  - one parameter *k*, in the case of *N* observations *N/k* paramters
  - high variance (not robust)
  - low bias

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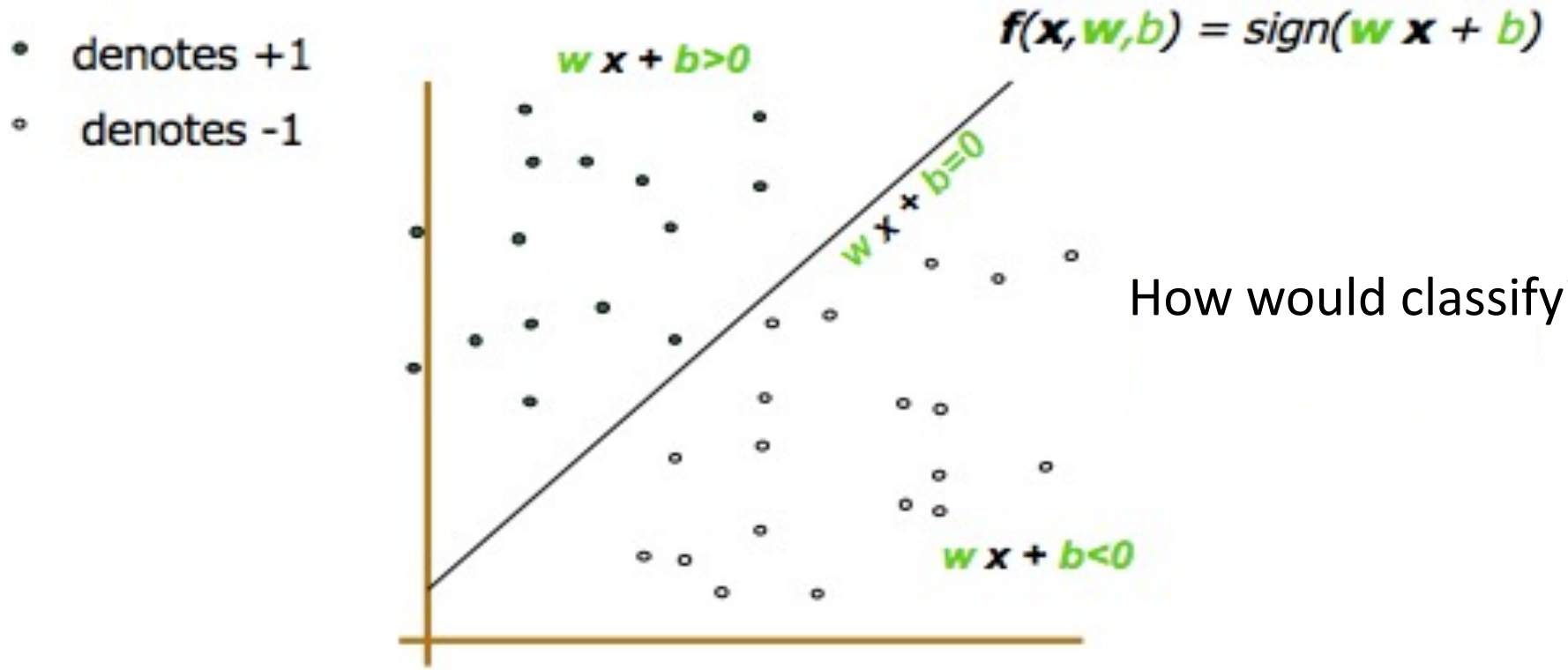
Binary classification problem



How would classify?

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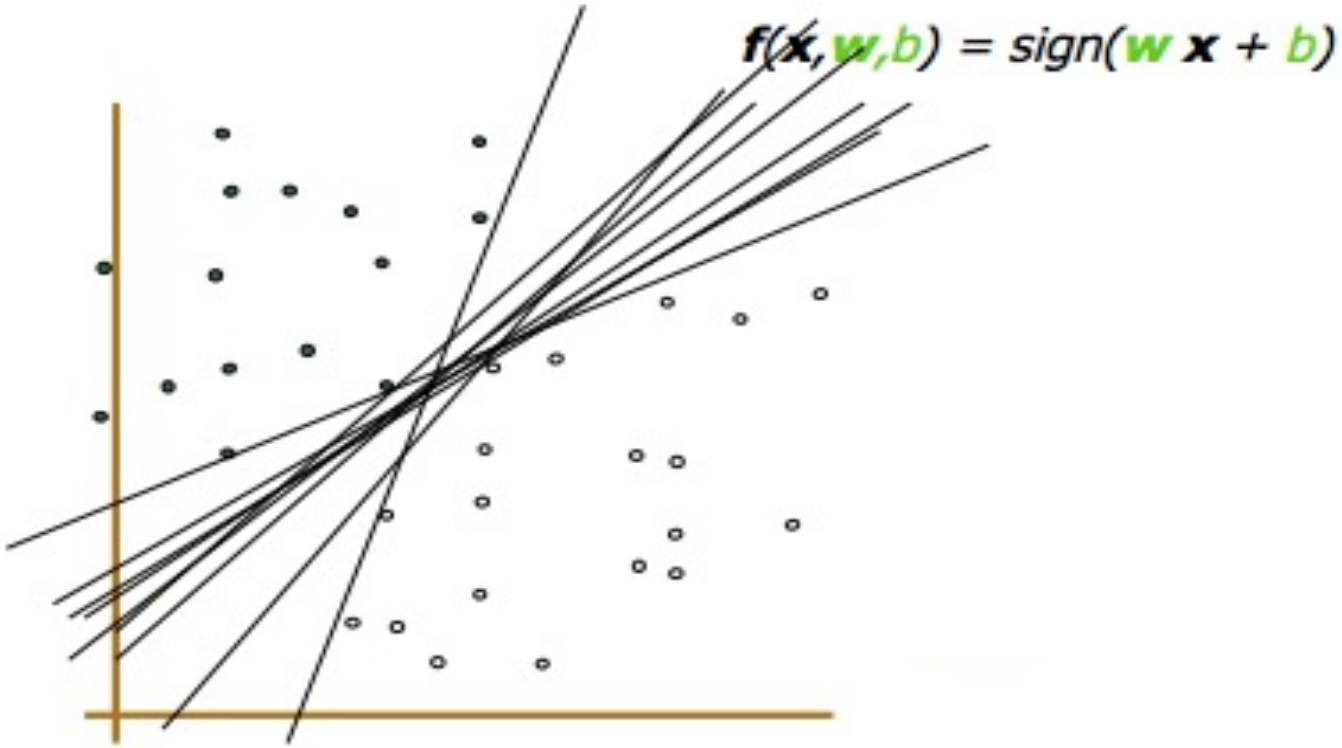
Binary classification problem (cont.)



There are more solutions. Which is the best?

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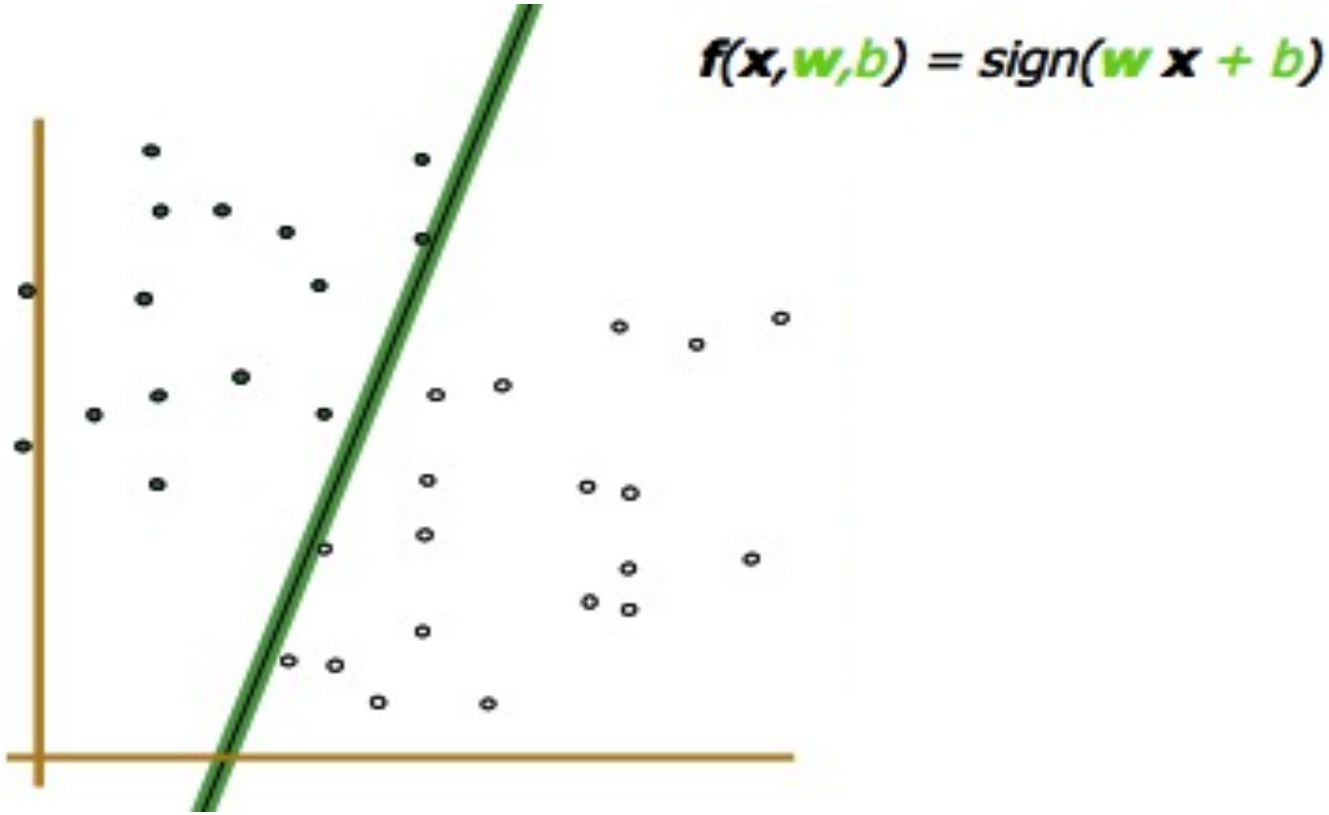
Binary classification problem (cont.)

Define the **margin** of a linear classifier as the width that the boundary could be increased by before hitting a datapoint



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Mathematical background of SVM

Goal

- Correctly classify all training data

$$wx_i + b \geq 1, \text{ if } y_i = +1$$
$$wx_i + b \leq 1, \text{ if } y_i = -1$$

- Maximize the margin

$$M = \frac{2}{\|w\|}$$

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Mathematical background of SVM (cont.)

We need to solve for *w* and *b* the following **quadratic optimization problem:**

Minimize

$$\phi(w) = \frac{1}{2} w^T \mathrm{w}$$

Subject to

$$y_i(wx_i + \mathrm{b}) \geq 1, \text{ for all } i$$

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Dataset with noise

- **Hard margin technique**: So far we require all data points be classified correctly. No training error.
- What if the training set is noisy? Solution: Use very powerful **kernel**s. Sometimes we need to face to **overfitting**.

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Soft margin technique

- Slack variables ξ_i can be added to allow misclassification of difficult or noisy examples.
- In this case our quadratic optimization criterion is

$$\phi(w) = \frac{1}{2} w^T \mathrm{w} + \mathrm{c} \sum_{i=1}^{k} \xi_i .$$

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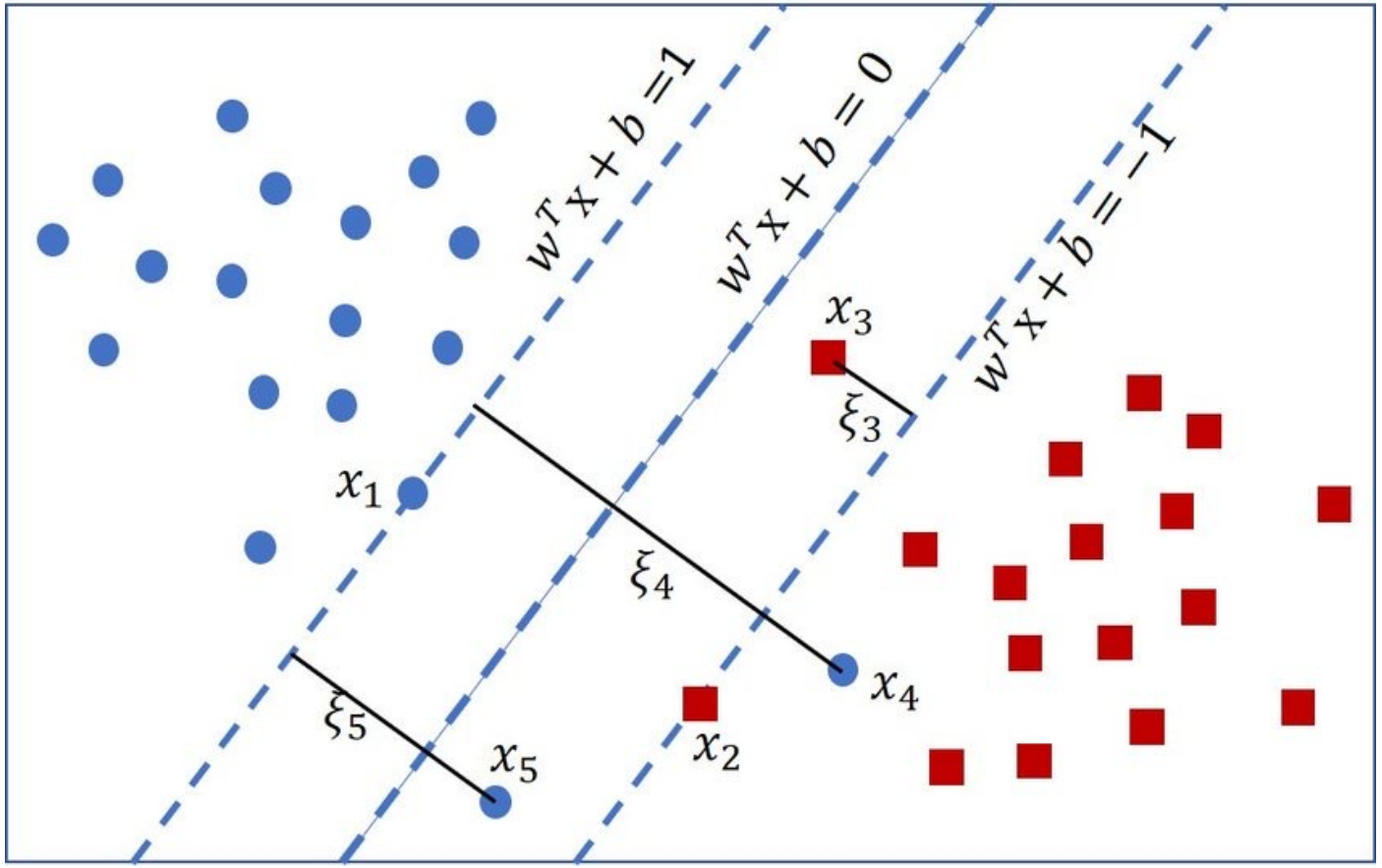
Soft margin classification



Source: https://www.researchgate.net/figure/SVM-with-soft-margin-kernel-with-different-cases-of-slack-variables_fig2_327015448

Non-linear SVMs

- Datasets that are linearly separable with some noises work out great.
- But what we can do if the dataset is just too hard?
- Let us map the original data into a higher-dimensional space where it is linearly separable.
- This mapping is based on the so-called **kernel function**.

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Examples of kernel functions

Linear: $K(\mathbf{x}_i,\mathbf{x}_j)= \mathbf{x}_i^T\mathbf{x}_j$

Polynomial of power p: $K(\mathbf{x}_i,\mathbf{x}_j)= (1+ \mathbf{x}_i^T\mathbf{x}_j)^p$

Gaussian (radial-basis function network):

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{2\sigma^2}\right)$$

Sigmoid: $K(\mathbf{x}_i,\mathbf{x}_j)= \tanh(\beta_0\mathbf{x}_i^T\mathbf{x}_j + \beta_1)$

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Properties of SVM

- Flexibility in choosing a similarity function.
- Sparseness of solution when dealing with large data set.
- Ability to handle large feature spaces, because the complexity does not depend on the dimensionality of the feature space.
- Overfitting can be controlled by soft margin approach.
- Nice math property: a simple convex optimization problem which is guaranteed to converge to a single global solution.
- Feature selection.

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Thank you for your attention